MAINTENANCE MODULE (FACILITIES & ASSET MANAGEMENT ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Maintenance Staff • Teachers • IT Team
Related Modules: Inventory • Finance • Notifications • Security • Analytics • Asset Management

---

1. MODULE OVERVIEW

1.1 Purpose

The Maintenance Module manages all facility-related operations within the school, including issue reporting, asset maintenance, and repair workflows. It enables:

- Reporting and tracking maintenance requests
- Assigning and managing maintenance tasks
- Scheduling preventive maintenance
- Monitoring facility conditions and assets
- Generating reports and analytics on maintenance activities

---

2. CORE OBJECTIVES

1. Provide a centralized system for reporting and tracking maintenance issues
2. Enable efficient task assignment and accountability
3. Support preventive maintenance scheduling
4. Track asset conditions and maintenance history
5. Improve response times and operational efficiency
6. Integrate with finance and inventory systems for cost tracking
7. Provide analytics for decision-making and planning
8. Ensure safety and compliance with facility standards

---

3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Issue Reported → Categorized → Assigned → In Progress → Resolved → Verified → Closed → Report Generated

3.2 Daily Operational Flow

User reports issue (e.g., broken desk)
↓
System categorizes and assigns priority
↓
Maintenance staff assigned task
↓
Staff updates progress and logs actions
↓
Issue resolved and verified
↓
Notifications sent to requester
↓
Data recorded for analytics and reporting

---

4. MODULE ARCHITECTURE

```
Maintenance Module
│
├── Request Management
├── Task Assignment & Workflow
├── Preventive Maintenance Scheduling
├── Asset & Equipment Management
├── Notifications & Communication
├── Analytics & Reporting
├── Mobile & Field Operations
├── Integration Layer (Finance/Inventory)
├── Safety & Compliance
├── Security & Role-Based Access
```

---

5. REQUEST MANAGEMENT

5.1 Features

- Submit maintenance requests
- Categorize issues (electrical, plumbing, structural, etc.)
- Assign priority levels (low, medium, high)
- Track request status (pending, in-progress, completed)

5.2 Data Model

Request

- request_id
- title
- description
- category
- priority
- status
- reported_by
- created_at

---

6. TASK ASSIGNMENT & WORKFLOW

6.1 Features

- Assign requests to maintenance staff
- Track task ownership and accountability
- Update task progress and completion status
- Maintain activity logs and comments

6.2 Data Model

Task

- task_id
- request_id
- assigned_to
- status
- start_date
- completion_date
- notes

---

7. PREVENTIVE MAINTENANCE SCHEDULING

7.1 Features

- Schedule recurring maintenance tasks
- Create checklists for inspections
- Automated reminders for upcoming maintenance
- Reduce unexpected breakdowns

---

8. ASSET & EQUIPMENT MANAGEMENT

8.1 Features

- Track assets (furniture, equipment, infrastructure)
- Maintain service history per asset
- Monitor asset condition and lifecycle

8.2 Data Model

Asset

- asset_id
- name
- category
- location
- condition
- last_maintenance_date

---

9. NOTIFICATIONS & COMMUNICATION

9.1 Features

- Alerts for new requests and assignments
- Notifications for overdue or completed tasks
- In-app messaging or comments within requests

---

10. ANALYTICS & REPORTING

10.1 Features

- Resolution rate tracking
- Average response and completion time
- Cost analysis by category
- Staff performance metrics
- Recurring issue identification

10.2 Dashboards

- Maintenance activity overview
- Issue trends by category
- Performance and efficiency reports

---

11. MOBILE & FIELD OPERATIONS

11.1 Features

- Mobile app for reporting and updating tasks
- Photo uploads for issue documentation
- Offline support with synchronization

---

12. INTEGRATION LAYER

12.1 Features

- Integration with inventory for spare parts tracking
- Integration with finance for cost management
- Link with asset management systems

---

13. SAFETY & COMPLIANCE

13.1 Features

- Compliance tracking (fire safety, lab safety, inspections)
- Escalation workflows for critical issues
- Audit trails for safety checks

---

14. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full control, reporting, configuration
Maintenance Staff	Manage assigned tasks
Teacher/Staff	Report issues, track requests
Auditor	View reports and compliance logs

14.2 Security Features

- Secure authentication and access control
- Audit logs for all maintenance activities
- Data protection and system integrity

---

15. SCALABILITY

- Support multi-campus deployments
- Handle high volumes of maintenance requests
- Cloud-based infrastructure with backup

---

16. IMPLEMENTATION GAPS (CRITICAL FIXES)

16.1 Current Issues Identified

- Limited role-based access and accountability
- No preventive maintenance scheduling
- Weak communication and notification system
- Limited analytics and reporting depth
- No integration with finance and inventory systems
- Lack of mobile support for field operations

16.2 Required Fixes

- Implement role-based task assignment
- Add preventive maintenance scheduling
- Enable notifications and communication features
- Build advanced analytics dashboards
- Integrate with finance and inventory modules
- Develop mobile-friendly interface

---

17. FUTURE ENHANCEMENTS

- AI-based predictive maintenance
- IoT integration for real-time equipment monitoring
- Automated cost optimization
- Smart asset lifecycle management
- Voice-enabled reporting for field staff